NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOTE: This drawing was approved by joint action of the Air Force and Navy Departments as the Air Force-Navy standard for this product. This drawing supersedes all antecedent standard drawings for the same product and shall become effective for the procurement of conventional supplies, or for use in new designs, not later than 6 months after the latest date of approval shown.

ACCESSORY CLEARANCE AT MOUNTING HOLES 1.500 PREFERRED

1.125 MIN

1.612 MAXIMUM (RETRACTED)

1.500 MAXIMUM (STARTER CONFIGURATION)

.310 ±.075

2.984 MAXIMUM

5.000 MAXIMUM FLANGE

(a)

.020 MAX RAD

.170 MAX
MAINTAIN .110 MIN PILOT LENGTH TO 2.997 PILOT DIA TOLERANCE

(1) 6, 12, OR 24 .344 +.010 −.005 DIAMETER HOLES EQUALLY SPACED WITHIN .005 OF BASIC POSITION IN RELATION TO 2.997 PILOT DIAMETER AND VERTICAL ℄

4.000 BASIC

.328 MIN

1.250 MINIMUM RADIUS WRENCH CLEARANCE

(1) PILOT DIAMETER 2.997 +.000 −.003

3 RADIAL TEETH EQUALLY SPACED WITHIN .002 AT 2.250 DIAMETER

LIGHTENING CUTS PERMISSIBLE TO SUIT DESIGN

(1) 2.250 MAX

1.000 THIS DIAMETER SHALL BE UNOBSTRUCTED TO .156 MIN DEPTH

STARTER JAW

(1) FLAT TOP OPTIONAL

.051 RAD

.031 RAD

.281 REF

.245 MIN

M

(1) OPTIONAL ROOT RELIEF. WIDTH MAY BE UNIFORM FOR LENGTH OF TOOTH FACE, BLENDING RADII PERMISSIBLE

.380 MAX

DEVELOPMENT OF TEETH AT 2.250 DIA
LEAD = .865 INCH PER REVOLUTION
(1) REFERENCE ANGLE M = 7°[illegible]

TYPE XIX-A DRIVE

(1) OPTIONAL ROOT RELIEF. WIDTH MAY BE UNIFORM FOR LENGTH OF TOOTH FACE, BLENDING RADII PERMISSIBLE

(1) 2.250 MAX

12 RADIAL TEETH EQUALLY SPACED WITHIN .003 AT 2.250 DIAMETER

.074

.031 RAD

.218 ±.005

M

THIS DIMENSION MAY BE PARALLEL AT ROOT, BLENDING RADII PERMISSIBLE

.074

1.625 THIS DIA SHALL BE UNOBSTRUCTED TO .125 MIN DEPTH

(1) DEVELOPMENT OF TEETH AT 2.250 DIA
LEAD = 3.500 INCHES PER REVOLUTION (3.4841 THEOR)
(1) REFERENCE ANGLE M = 26°20'22"

STARTER JAW

TYPE XIX-B DRIVE

(1) (a) ROTATION: WILL BE AS VIEWED FROM DIRECTION SHOWN.

ACCESSORY SHALL BE CAPABLE OF SATISFACTORY OPERATION WITH THE GEOMETRIC CENTER OF THE DRIVING JAW DISPLACED TO A MINIMUM OF .010 IN ANY DIRECTION FROM THE CENTER OF PILOT DIAMETER 2.997.

FLANGE OUTLINE AS DIMENSIONED IS A BASIC MAXIMUM AREA REQUIREMENT.

STARTER JAW TOOTH SURFACE HARDNESS SHALL BE ROCKWELL 055 MAXIMUM.

REMOVE ALL BURRS AND SHARP EDGES.

FOR DETAIL OF TYPE XIX ENGINE DRIVE, SEE AND20009.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

(1) THIS DRAWING AND THE APPLICABLE ACCESSORY SPECIFICATIONS, TOGETHER, COMPLETELY DEFINE THE DESIGN REQUIREMENTS.

NORMAL USE OF THIS FLANGE IS FOR STARTERS FOR RECIPROCATING ENGINES.

(1) WHEN SPECIFIED BY THE ACCESSORY SPECIFICATION, DRAIN BOSS(ES) WITHIN THE FLANGE OD SHALL BE PROVIDED IN ACCORDANCE WITH AND10050 FOR A .250 TUBING WITH PLUG AN814-4 AND GASKET AN6290-4 SIZE TO ALLOW ONE BOSS TO BE WITHIN 30° OF THE BOTTOM POSITION WHEN THE TERMINAL BLOCK IS MOUNTED IN ANY POSITION.

APPROVED 1 Aug 49 REVISED (1) 4 Mar 53

AIR FORCE-NAVY AERONAUTICAL DESIGN STANDARD

FLANGE – TYPE XIX ACCESSORY MOUNTING

AND10269

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